

HARVEE AGRAVANTE

Technical Service Manager · IT Systems & Infrastructure · Bangkok, Thailand

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PROFILE

Systems manager and full-stack engineer, 8 years inside Thai international schools. I run the production ERP at Oneness Bilingual School — *erp.oneness.ac.th* — alone, across 3 campuses and 18 roles. Stood up Aruba Central and the staff/student ID system at Beaconhouse Yamsaard Ladprao, ran Eskooly Pro rollouts for two other schools concurrently, and contracted IT for Junior Achievement Thailand. Comfortable owning a stack from VPS up, working solo, and coordinating Thai vendors and leadership. I write the runbook because I'm the one paged at 2 a.m.

CORE COMPETENCIES

Networking	Aruba Central (cloud-managed), VLAN segmentation, per-profile bandwidth caps, content filtering, AP & switch deployment.
Service delivery	Multi-site IT ops, SLA accountability, vendor coordination, procurement, commissioning, handover docs.
Systems & infra	Self-managed VPS, PM2, Cloudflare, Redis, BullMQ, MinIO (S3-compatible), cron-driven backups, Docker.
Identity & access	Unified employee/student ID issuance, RBAC, JWT auth, TOTP 2FA, role-scoped access control.
Security	AES-256-GCM field encryption, Argon2id, append-only audit hash chains, CSP, threat modeling, DR procedures.
Attendance systems	ZKBio Zlink (ZKTeco facial terminals), shift/timetable design, BEST-protocol device onboarding.
SIS / school platforms	Eskooly Pro deployment & admin, Google Workspace for Education, Microsoft 365 A1.
Curriculum design	Computer Science scope-and-sequence (P1-P6), EY unplugged coding (Code.org CS Fundamentals), bilingual translation.
Development stack	Next.js 15, TypeScript, React 19, Prisma, PostgreSQL, Redis, BullMQ, Socket.io, Tailwind, Express, Apps Script.
Languages	English (C2) · German (B1) · Spanish (elementary) · Thai (working comprehension).

PROFESSIONAL EXPERIENCE

IT Systems Manager & Full-Stack Engineer · Oneness Bilingual School · Bangkok *Apr 2026 – Present*

Sole administrator of the Oneness ERP — multi-campus K-12 platform across NTB-01 Nonthaburi, NKP-01 Nakhon Pathom, PTM-01 Pathumthani, plus head office. 18 roles, 157 Prisma models. Stack: Next.js 15 / TypeScript / Prisma / Postgres / Redis / BullMQ / Socket.io. Live at *erp.oneness.ac.th*.

- Built the agent layer that keeps role data clean — 5 TypeScript modules under *src/lib/agents/* (role-registry, types, data-integrity, onboarding, structure-redundancy) plus a one-shot migration that renamed the legacy "planning" role to *head_of_academics* across the live database without downtime.
- Ran a security audit via a self-written *erp-audit.mjs* script — 12 checks covering CORS, rate limiting, JWT verification, Zod validation, raw-SQL injection risk, Prisma index coverage, .env hygiene, structured logging. Closed three critical gaps before they reached parents.

- Integrated ZKBio Zlink facial attendance for staff across the 3 campuses. Traced a "no late flags" failure to 3 compounding causes — Check-In Period After Work set to 0 minutes, Calculate engine never run, gate device provisioned as Access Device instead of Time Attendance terminal.
- Designed the OBS Computer Science scope-and-sequence — Primary 1 to Primary 6 course outlines, syllabi, and lesson plans, plus the Early Years unplugged coding programme (K2–K3, mapped to Code.org CS Fundamentals Course 1, 18 print-ready worksheets).
- Wrote the 60-page Oneness ERP Staff Handbook v2.1 — role workflows, security posture, VPS runbook, DR procedure. Single SOP across all three campuses.
- Owned the deployment pipeline on a self-managed VPS: PM2 supervision, Cloudflare for TLS and edge caching, MinIO object storage, Redis for sessions and BullMQ, cron-driven Postgres backups with offsite copies.
- Built a parent-contact registration pipeline for facial-scan attendance notifications — Google Apps Script generates the form programmatically (Thai mobile-number and student-number validation), responses land in a linked sheet, *onFormSubmit* handler queued to hit the ERP invite endpoint.
- Public artifacts: handbook at kimharvee.github.io · 5-minute production walkthrough at youtu.be/I-XD-yMaFyY.

Information Coordinator (IT) · Beaconhouse Yamsaard School–Ladprao · Bangkok *Feb 2025 – Mar 2026*

- Stood up the campus network on Aruba Central — cloud-managed switches and access points (Aruba AP-725), VLAN split between staff and student traffic, bandwidth caps and content filtering per profile. Owned requirements, vendor negotiation, configuration, commissioning, handover.
- Designed and rolled out the unified employee + student ID system across the campus.
- Got Google Workspace for Education and Microsoft 365 A1 approved for the school — both free at the education tier.
- Administered user accounts and edtech platforms; ran integration training for teaching staff. Saturday coding sessions using Roblox Education and Code.org.

Concurrent IT & Curriculum Contracts · LittleGaia · BIS · Junior Achievement Thailand *Feb 2025 – Mar 2026*

- Deployed and configured Eskooly Pro (school management SIS) for LittleGaia and BIS — tenant setup, user provisioning, academic structure modelling, staff training.
- Part-time IT support for Junior Achievement Thailand alongside the Beaconhouse role.
- Designed and translated the Quirky curriculum (English ↔ Thai) for partner schools.

Multidisciplinary Science Educator · Sisaket Wittayalai School · Sisaket *May 2024 – Feb 2025*

- Biology, Chemistry, Geology, and Astronomy in English across multiple year groups. Integrated AI tooling (Magic School, task-specific custom GPTs) into delivery and assessment.

Earlier educator roles · **Biology, Chemistry, English** · Darasamutr Sri Racha · Mathayom Watmai Krongtong · St. Michael's Academy · Culipapa NHS (PH) *2017 – 2024*

EDUCATION & CREDENTIALS

MA in Education, Biology Teaching · Marinduque State University, Philippines

BSEd, Biological Sciences · University of Negros Occidental — Recoletos, Philippines

Programming & CS foundations · Codecademy, freeCodeCamp, self-directed (ongoing)

Professional Teaching License (Philippines, PRC) · Microsoft STEM Education Certified Educator · Code.org, CodeHS, Magic School AI — Certified Teacher

Publication · Agravante, H. T. & Labay, P. M. (2026). "Spiral Progression Approach in English-Medium Science Instruction." *Psychology and Education: A Multidisciplinary Journal (PEMJ)*, Vol. 54(4). Scopus-indexed.